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Radiation

1) Define Radiation heat transfer
2) Define the term absorptivity, reflectivity and transmittivity of radiation
3) State the following law:
   i) Wien's displacement law ii) Stefan Boltzmann law
4) State the following law:
   i) Krichhoff's law ii) Planck's law
5) What is black body ? How is differ from a gray body ?
6) The effective temperature of the body having an area of 0.12 m^2 is 527 ^oC.
   Calculate the following
   i) The total rate of energy emission
   ii) The wave length of maximum monochromatic emissive power
7) Calculate the following for an industrial furnace in the form of a black body and emitting radiation at 2500 ^oC.
   ii) Monochromatic emissive power at 1.2 μm length
   iii) Wave length at which the emissive is maximum
   iv) Maximum emissive power
   v) Total emissive power
   vi) Total emissive power of the furnace if its assumed as a real surface with emissivity equal to 0.9
8) Define Shape factor and mention salient features of its.
9) Mention the shape factor's fact and properties for specific geometries and for the analysis of radiant relation exchange between surfaces.